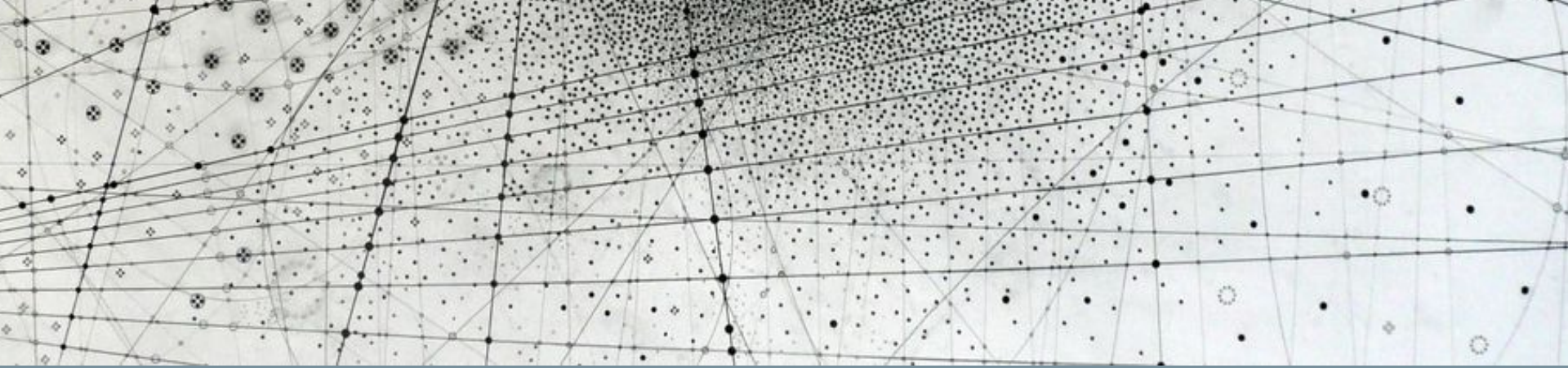


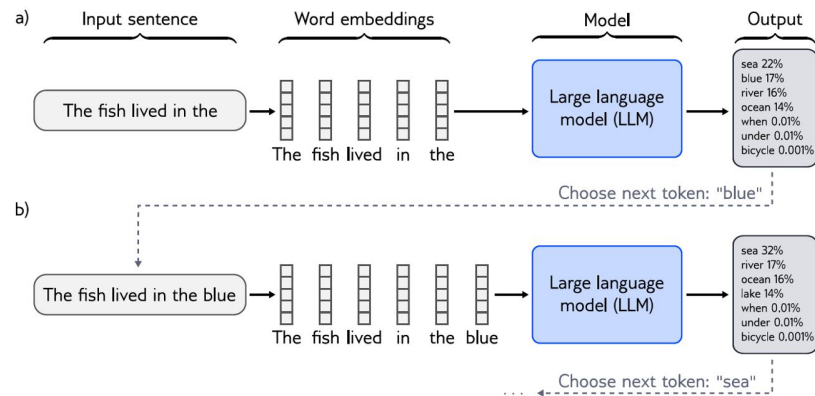
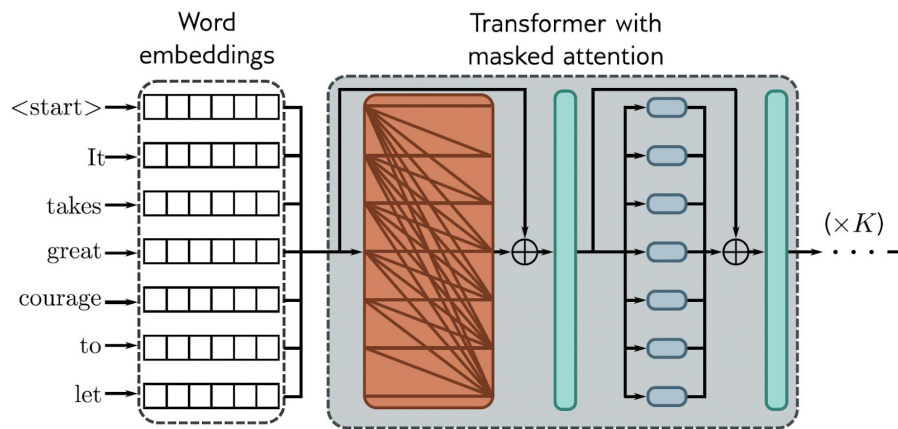


Research Methods in Machine Learning & Physics

Quick recap

LLMs

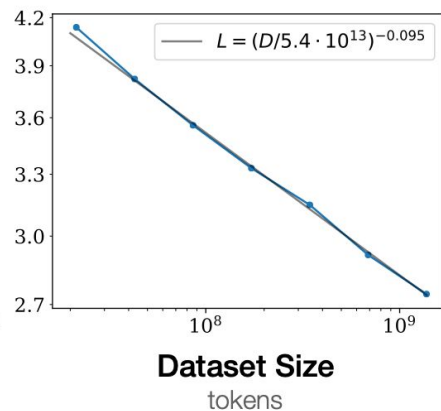
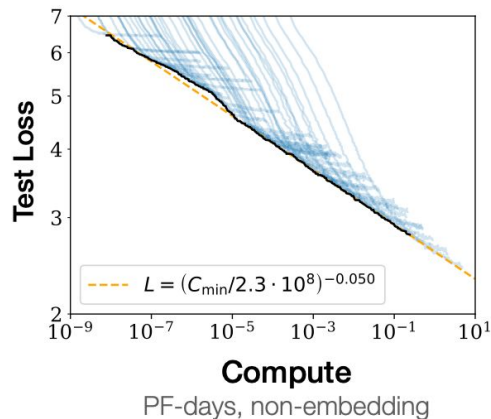
Decoder-only framework → next-token prediction → generative!



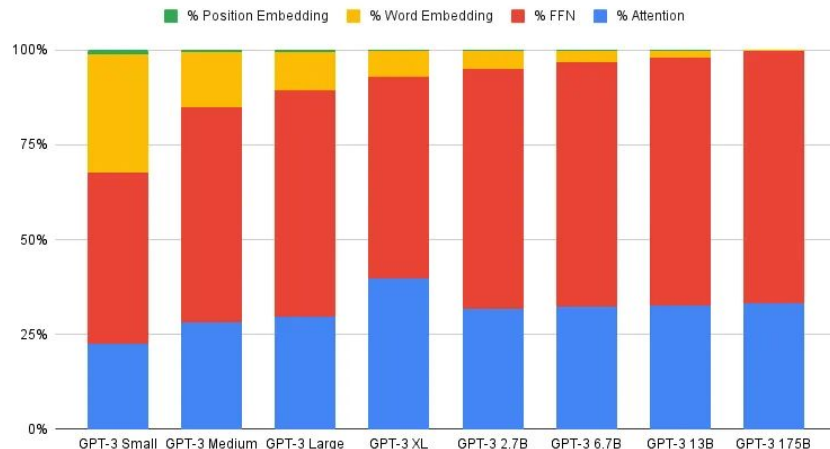
LLMs

Massive #s of tokens in the training data & model parameters due to observed scaling laws in performance.

Interestingly, the majority of model parameters come from the MLP layers, not the attention layers.



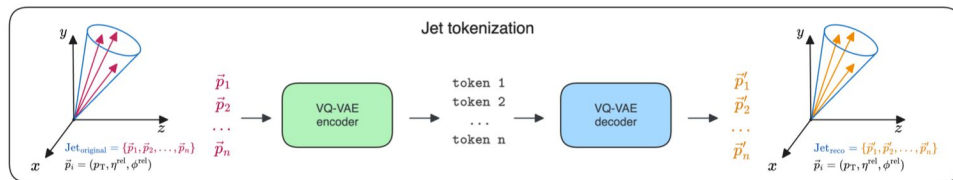
GPT-3 Parameter Usage



About 1/3 of What
Attention is ~~All~~ You Need

LLMs for Physics

1. Autoregressive next-token prediction is a useful framework beyond language
 - a. e.g. predict the “next particle” in a jet:



2. Scaling laws are critical in scientific ML, too
3. Actually, maybe language *is* relevant for physics analyses
 - a. Either as an extra modality (e.g. papers)
4. Using LLMs for multimodal embeddings & data analysis
5. LLMs for physics theory / proof-solving / interpretability

Self-Supervised Learning (SSL)

Representation learning

Representation learning = learning representations of the data that make it easier to extract useful information when building classifiers or other predictors

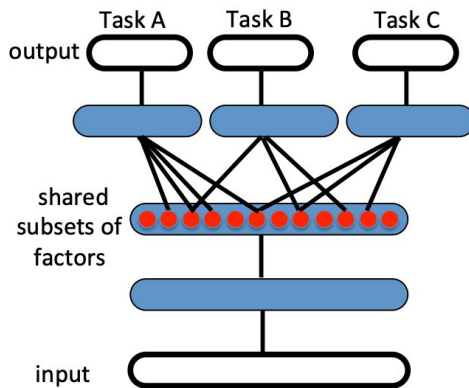


Fig. 1. Illustration of representation-learning discovering explanatory factors (middle hidden layer, in red), some explaining the input (semi-supervised setting), and some explaining target for each task. Because these subsets overlap, sharing of statistical strength helps generalization.

Self-supervised learning (SSL)

Self-supervised learning (SSL) is a specific approach to representation learning in which you empower a model to define its own labels.

This allows for training on large unlabeled datasets, but it's not exactly “unsupervised” – rather, the model performs a “pretext” task designed to help it learn essential features in the data.

(Note: the autoencoder is therefore a kind of SSL!)

Why care about SSL?

- SSL only recently emerged in the last ~5 years or so, but it has since exploded in popularity
- It is the backbone of most large-scale foundation models, e.g. LLMs and beyond
- Learning useful representations from unlabeled data allows for lots of possibilities

Self-supervised learning (SSL)

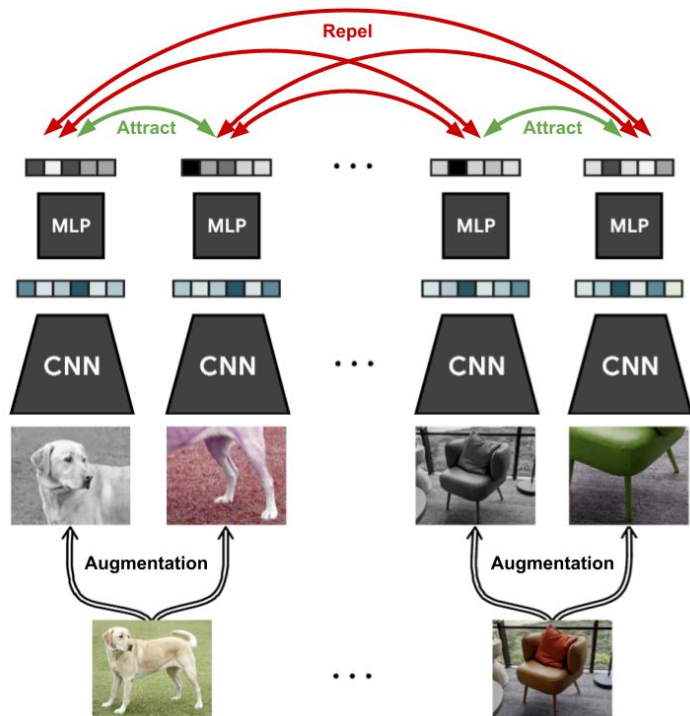
Types of SSL include **contrastive** & **non-contrastive** (e.g. generative).

Contrastive SSL means comparing pairs of examples at a time and learning to bring similar ones together and push different ones apart.



Self-supervised learning (SSL)

For today, we will focus on one canonical example of **contrastive SSL** called **SimCLR**.



v3 [cs.LG] 1 Jul 2020

A Simple Framework for Contrastive Learning of Visual Representations

Ting Chen¹ Simon Kornblith¹ Mohammad Norouzi¹ Geoffrey Hinton¹

Abstract

This paper presents *SimCLR*: a simple framework for contrastive learning of visual representations. We simplify recently proposed contrastive self-supervised learning algorithms without requiring specialized architectures or a memory bank. In order to understand what enables the contrastive prediction tasks to learn useful representations, we systematically study the major components of our framework. We show that (1) composition of data augmentations plays a critical role in defining effective predictive tasks, (2) introducing a learnable nonlinear transformation between the representation and the contrastive loss substantially improves the quality of the learned representations, and (3) contrastive learning benefits from larger batch sizes and more training steps compared to supervised learning. By combining these findings, we are able to considerably outperform previous

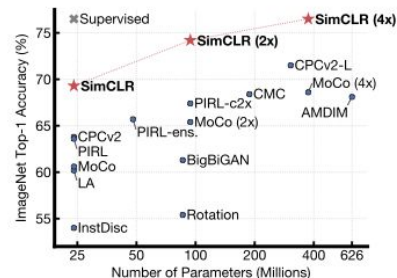


Figure 1. ImageNet Top-1 accuracy of linear classifiers trained on representations learned with different self-supervised methods (pretrained on ImageNet). Gray cross indicates supervised ResNet-50. Our method, SimCLR, is shown in bold.

In-class work

One person from each group will present their solution to the class:

Task 1: Explain the big-picture SimCLR methodology on the chalkboard

Task 2: Implement a DataLoader in PyTorch to produce 2 augmentations of an input image on a dataset of your choice

Task 3: Explain the InfoNCE loss on the chalkboard